

DMS Short Notes (Quick Revision)

Unit 1

Set - Collection of well defined element.

↓ (form)
Set builder notation $A = \{x : \text{Cond.}\}$
Roster $A = \{1, 2, 3\}$

Types → empty set (ϕ)
finite set
Infinite set
equal set $A = B$
equivalent set $n(A) = n(B)$
subset $A \subseteq B$
proper subset $A \subset B$

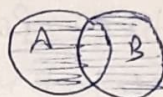
Power set $n(A) = a$

$$P(A) = 2^a$$

ϕ & A is also subset of A .

operation

$A \cup B$
union



$A - B$



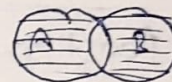
$$A - (A \cap B)$$

$A \cap B$

intersection



$A \oplus B$



Symmetric

$$(A - B) \cup (B - A)$$

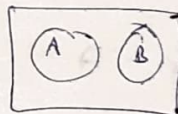
complement

$$A' = U - A$$



Disjoint

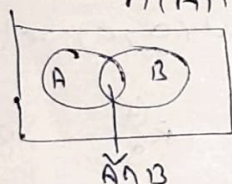
$$A \cap B = \phi$$



Principle of inclusion & exclusion

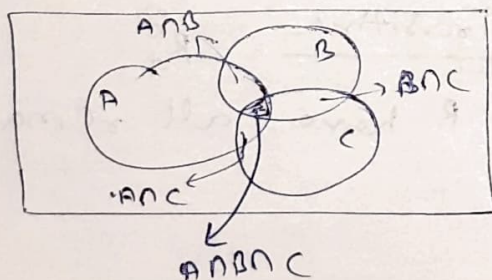
$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

for A, B, C



$$A(A \cap B)$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$



Property

1. Associative

$$A \cup (B \cap C) = (A \cup B) \cap C$$

2. Distributive

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

3. Commutative

$$A \cap B = B \cap A$$

* Relation - Subset of all ordered pair
 $R \subseteq A \times B$

(ordered pair)
 (a, b)
 $a \in A$
 $b \in B$

Cartesian product

$A \times B =$ set of all ordered pair

If Relation is

Reflexive + Symmetric +
 Transitive $\Rightarrow$ equivalent

Reflexive + Antisymmetric +
 Transitive $\Rightarrow$ P.O.S.E.T
 (partial order relation)

Types

Reflexive

$$\forall a \in A$$

$$(a, a) \in R$$

Symmetric

$$\forall a, b \in A$$

$$(a, b) \in R \Rightarrow (b, a) \in R$$

$$(1, 2) \in R \Rightarrow (2, 1) \in R \checkmark$$

Transitive

$$\forall a, b, c \in A$$

$$(a, b) \in R, (b, c) \in R$$

$$\text{So } (a, c) \in R \checkmark \text{ (must)}$$

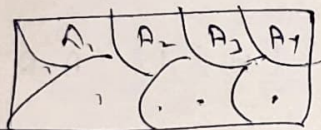
Antisymmetric

$$\forall a, b \in A$$

$$\text{if } (a, b) = (b, a)$$

$$\text{then } a = b$$

Partition of set



$$\forall A_i \cap A_j = \emptyset$$

$$A_1 \cup A_2 \cup A_3 \cup A_4 = A$$

Close Relation

fulfill the requirements

1. Reflexive

$$R_{\Delta}^{\oplus} = R \cup \Delta$$

$$\Delta = \text{all } (a, a) \text{ element}$$

2. Symmetric

$$R_S^{\oplus} = R \cup R^{-1}$$

$$(a, b) \in R \Rightarrow (b, a) \in R^{-1}$$

(3)

Transitive

$$R_T^{\oplus}$$

R have all remaining part

* Function Special case of relation

$$f: A \rightarrow B$$

$\forall A$, image in B

all image are unique.

means $\forall f(x)$, B is available.

Domain A

Codomain

dom, range

$$f(A) \subseteq B$$

Polynomial $\Rightarrow$ Real numbers

$$f(x) = \frac{g(x)}{h(x)} \Rightarrow h(x) \neq 0$$

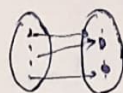
Types

$\rightarrow$ one-one - Injective $\Rightarrow f(a) = f(b) \Rightarrow a = b$

$\rightarrow$ on-to - surjective

many-one

$$f(a) = f(b) \Rightarrow a \neq b$$

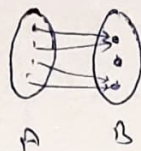


Single image of A in B
Codomain = Range, for all B image in A .

In-to

multiple image of B in A .

(multiple value give same answer)



Bijjective

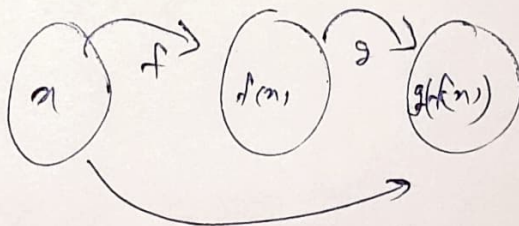
Inverse function

Bijjective (injective + surjective)

one-one + on-to.

$$f: A \rightarrow B \Rightarrow f^{-1}: B \rightarrow A$$

Composite function



$$f: A \rightarrow B$$

$$g: B \rightarrow C$$

$$g \circ f: A \rightarrow C$$

$$f \circ g: C \rightarrow A$$

$$f \circ g(x) = f(g(x))$$

$$g \circ f(x) = g(f(x))$$

Mathematical Induction

Steps $n = 1$ put

$n = k$ let it is true

$n = k+1$ prove with help of

$n = k$ property

* Region Hall Principal

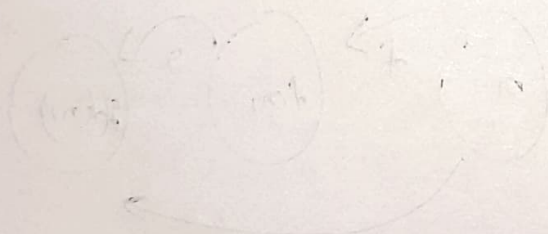
In n region

m region hall

& $n > m$

then 1 region hall with more than 1 region.

~~Let~~



Unit-2

(Propositional logic)

Propositional logic $\rightarrow$ Either true or false

Logical operators

Negation ($\neg$)

$$T \rightarrow F$$

$$F \rightarrow T$$

And (Conjunction)

$$P \wedge Q$$

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

OR (Disjunction)

$$P \vee Q$$

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Conditional

$$P \rightarrow Q \quad | \quad P \Rightarrow Q$$

P	Q	$P \rightarrow Q$
T	T	T
* T	F	F
F	T	T
F	F	T

Bi-conditional

(Same) like $\Rightarrow$ T

$$P \leftrightarrow Q$$

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Tautology

all True

Contradiction
(Fallacy)

all false.

Converse $\Rightarrow P \rightarrow Q \Rightarrow Q \rightarrow P$

Contrapositive $\Rightarrow \neg Q \rightarrow \neg P$

Inverse $\Rightarrow \neg P \rightarrow \neg Q$

properties

$(P \vee Q) = (Q \vee P) \Leftrightarrow (P \wedge Q) = (Q \wedge P)$ Commutative property

$(P \vee Q) \vee R = P \vee (Q \vee R) \Rightarrow$ Associative

$P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R) \Rightarrow$ Distributive

$\neg(P \vee Q) = \neg P \wedge \neg Q$ DeMorgan's.

$P \vee (P \wedge Q) = P$ absorption.

$P \vee T = P$ Identity.

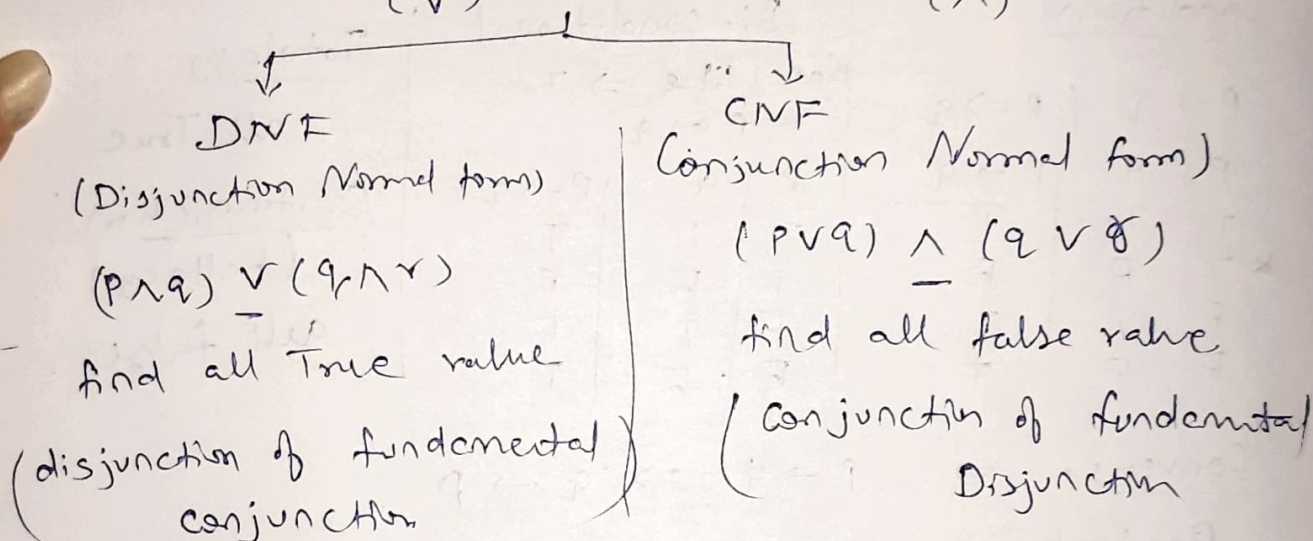
Predicates $P(n)$ predicate is a sentence that will be true / false.

Quantifiers $\rightarrow$ on which value predicate will be true.

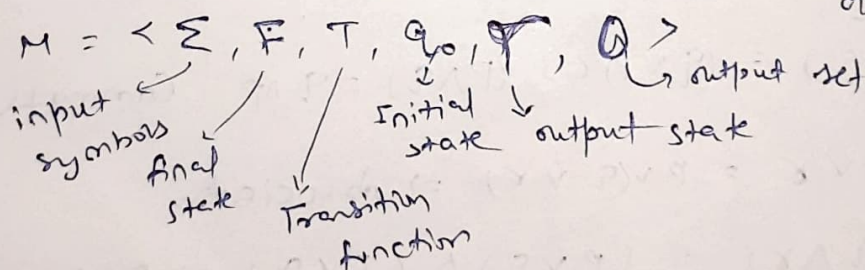
Existence ($\exists$) $\rightarrow$ for at least one element of n in D will be true.
 Universal ($\forall$) $\rightarrow$ for all elements of n in D will be true.

$$\forall n \in D, P(n)$$

Normal Form sum of minterm or product of maxterm
 ($\vee$) ($\wedge$)



Finite state Machine number of state finite & which take input & give output
 6 tuple.



$\Rightarrow$ Two machine are FSM, if they produce same output for every possible input string
 equivalence

Unit-3 Poset & Hasse diagram.

Poset (Partial order Relation)

- Reflexive $(a, a) \in R$
- Antisymmetric $(a, b) = (b, a) \rightarrow a = b$
- Transitive $(a, b), (b, c) \in R$
 $(a, c) \in R$

Hasse diagram

Successor
↑
Preceder

$$A = (\{a, b, c, d\}, \leq)$$

or $A = (A, \leq)$
Symbol.

maximal $\rightarrow$ Top element

minimal $\rightarrow$ Bottom element

Upper bound. $(S, \leq)$
(c) $(a, b) \in S$
 $b \leq c, a \leq c$

Lower bound $(S, \leq)$

(d) $(a, b) \in S$
 ~~$d \geq a, d \geq b$~~
 $d \leq a, d \leq b$

LUB
(least upper bound)

Supreme

GLB

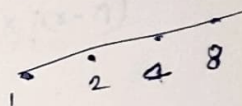
Greatest Lower bound.

(lowest)

$\rightarrow$ special elements

Total

(Total order set
Chain order set)
(linear)



all elements are comparable

Lattice consist LUB & GLB.

POSET

$a \vee b$ & $a \wedge b$ exist.
LUB $\hookrightarrow$ GLB.

Properties

1. Commutative
 $a \vee b \Rightarrow b \vee a$
 $a \wedge b \Rightarrow b \wedge a$

2. Distributive
 $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

3. Associative
 $a \vee (b \vee c) = (a \vee b) \vee c$

Absorption

$$\left. \begin{aligned} a \wedge (a \vee b) &= a \\ a \vee (a \wedge b) &= a \end{aligned} \right\}$$

Lattice Homomorphism

(i) Join

$$f(x \vee y) = f(x) \vee f(y)$$

(ii) Meet

$$f(x \wedge y) = f(x) \wedge f(y)$$

mapping of two Lattice

L & M ,

(iii) Order.

$$x \leq y$$

$$f(x) \leq f(y)$$

$$(f: L \rightarrow M)$$

Sub Lattice

$\rightarrow$ Non empty subset of Lattice.

$$a \vee b \in S$$

$$a \wedge b \in S$$

$$(L, \vee, \wedge)$$

$$S \subseteq L$$

Complete Lattice

Consisting LUB & GLB
($\vee$)
for every subset.

Bounded Lattice

Least element = 0

Greatest element = 1

$$(0 \leq a \leq 1)$$

$$(a \in L)$$

$$a \vee 0 = a$$

$$a \wedge 1 = a$$

Complemented Lattice

$$a \vee b = 1$$

$$a \wedge b = 0$$

Distributive lattice

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

Combination

$${}^nC_r = \frac{n!}{(n-r)!r!}$$

Permutation

$${}^nP_r = \frac{n!}{(n-r)!}$$

~~for repetitive letter keep it~~

Binomial Theorem

$$(x+y)^n = {}^nC_0 x^n y^0 + {}^nC_1 x^{n-1} y^1 + \dots + {}^nC_n x^0 y^n$$

$$= \sum_{r=0}^n {}^nC_r x^r y^{n-r}$$

Recurrence Relation :- Sequence of real number in which
prev. term is generate next term.

$$a_n = a_{n-1} - a_{n-2}$$

$$(n - (n-2) = 2 \text{ order})$$

order $\rightarrow$ difference b/w
highest & lowest subscript

* Recurrence Relation in Linear form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

Homogeneous

C1 roots same. $= r$

$$CF = a_n = (C_1 + n C_2) r^n$$

C2 ~~same~~ roots diff. r_1, r_2

$$a_n = (C_1 r_1^n + C_2 r_2^n)$$

C3 Imaginary Roots.

$$\alpha \pm i\beta$$

$$a_n = (\alpha^2 + \beta^2)^{n/2} (C_1 \cos n\theta + C_2 \sin n\theta)$$

$$\theta = \tan^{-1}\left(\frac{\beta}{\alpha}\right)$$

Non-Homogeneous

CF + PI

C1 if $f(x) = ax^2 + bx + c$

$$PI = p_2 x^2 + p_1 x + p_0$$

C2 $f(x) = a(b)^n$

$$PI = p_0(b)^n$$

C3 $f(x) = a n (b)^n$

$$PI = (p_1 n + p_0)(b)^n$$

* Generating Function (format power series) $A(x)$

$$A(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_r x^r + \dots$$

$$A(x) = \sum_{r=0}^{\infty} a_r x^r$$

Normal

$$a_r = 1$$

$$1 + x + x^2 + \dots \Rightarrow \frac{1}{1-x}$$

$$a_r = r$$

$$1 + x + 2x^2 + 3x^3 + \dots \Rightarrow \frac{x}{(1-x)^2}$$

$$a_r = r^2$$

$$x + 2x^2 + 3x^3 + \dots \Rightarrow \frac{x(x+1)}{(1-x)^3}$$

$$a_r = {}^n C_r$$

$${}^n C_0 x^0 + {}^n C_1 x^1 + \dots \Rightarrow (1+x)^n$$

$$a_r = \alpha^r$$

$$1 + \alpha x + \alpha^2 x^2 + \alpha^3 x^3 + \dots = \frac{1}{1-\alpha x}$$

Unit - 4 (Algebraic Structure)

Binary Operation

Set A let $a, b \in A$

$$A * A \rightarrow A$$

$$a * b \in A$$

$$(\mathbb{N}, +) \checkmark$$

$$(\mathbb{N}, \cdot) \checkmark$$

$$(\mathbb{N}, -) \times$$

$$(\mathbb{Z}, \cdot) \checkmark$$

$$(\mathbb{Z}, +) \checkmark$$

$$(\mathbb{Z}, -) \checkmark$$

Algebraic Structure

$$\{A, f_1, f_2, \dots\}$$

$$\Rightarrow (\mathbb{R}, +, \cdot)$$

$$(\mathbb{N}, +)$$

Binary operation

Properties

(1) Closure Property

$$a \in A$$

$$b \in A$$

$$a * b \in A$$

$$\forall a, b \in A$$

(2) Associative property

$$a * (b * c) =$$

$$(a * b) * c$$

existence of
(3) Identity

$$e * a = a$$

$$\forall a \in A$$

$$\text{add. } \Rightarrow$$

$$0 + a = a$$

$$\text{mul. } \Rightarrow$$

$$1 * a = a$$

(4) Existence of Inverse

$$a * \frac{1}{a} = e(1) \Rightarrow (.)$$

(5) Commutative

$$a * b = b * a$$

$$a * a = e(0) (+)$$

Semi Group

- Associative
- Closure

Sub semi group

- Subset of semigroup.
- non empty subset

Cyclic Semi group

$$n = a * a * a$$

a = generator which generates all terms.

Monoid

- Associative
- Closure
- Identity.

Submonoid

- subset of monoid
- non empty subset

Cyclic monoid

$$n = a * a * a \dots$$

a = generator which generates all terms

Group

- Associative
- Closure
- Identity
- inverse.

Subgroup

- subset of group.
- Non empty subset

Cyclic group

$$n = a * a * a \dots$$

a = generator

* Abelian Group a group with commutative.

- Associative
- Closure
- Inverse
- Identity.
- Commutative

Group Homomorphism

$$(G, *) , (G', *')$$

$\Rightarrow$ 2 group.

$f: G \rightarrow G'$ mapping

$$f(a * b) = f(a) *' f(b)$$

Normal Subgroup

$$gHg^{-1} = H \quad \forall g \in G$$

$$gH = Hg \quad \forall g \in G$$

$H = \text{subgroup} \quad H \leq G$

$G = \text{group}$

H is normal if it is
invariant under
conjugation

Group Isomorphism

- Homomorphism

- one-one

- on-to.

permutation group

one-one + on-to.
(bijective)

(non-abelian for $n \geq 3$)

rearrangement under composition

COSET

$H = \text{Subgroup}$

$a \in G$

Left

$$aH = \{ah \mid h \in H\}$$

Right

$$Ha = \{ha \mid h \in H\}$$

Factor Group (Quotient)

$$G/H = \{aH \mid a \in G\}$$

$$(aH)(bH) = (ab)H$$

$$\text{eg } G = \mathbb{Z} \quad H = 3\mathbb{Z}$$

$$G/H = \{0+H, 1+H, 2+H\}$$

mod 3

for mod 4

$$H = 4\mathbb{Z}$$

$$\left. \begin{array}{l} a \equiv b \\ 5 \equiv 1 \\ 6 \equiv 2 \\ 8 \equiv 0 \end{array} \right\} \begin{array}{l} \text{mod } 4 \\ \text{mod } 4 \\ \text{mod } 4 \end{array}$$

remainder same

Ring & Fields

Ring $(R, +, \cdot)$

$(R, +)$ abelian group.

$(R, \cdot) \rightarrow$ associative

Distributive law.

$$a(b+c) = ab+ac.$$

Note

zero is unique

- Additive inverse $\Rightarrow$ unique

$$a \cdot 0 = 0 \cdot a = 0$$

$$a(-b) = -(ab)$$

$$(-a)(-b) = ab$$

- Cancellation $\rightarrow$ left cancel
left only

right side cancel = right

$$a \cancel{b} a = a \cancel{b} a \quad \checkmark$$

$$\cancel{a} b a = a \cancel{b} a \quad \times$$

Types $\rightarrow$ Commutative Ring

$$ab = ba$$

Ring with Identity =

$$\exists 1 \Rightarrow a \cdot 1 = a$$

Integral Domain.

(commutative with no zero
division)

Zero division $ab = 0$.

$$a \neq 0, b \neq 0$$

Field

$$(F, +, \cdot)$$

$$(F, +) = \text{abelian}$$

$$(F, \cdot) = \text{abelian}$$

non zero element

$$(F, \cdot) = \text{Commutative}$$

Unit - 5

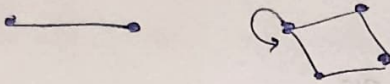
(Graph Theory)

Types

Graph written as $G(V, E)$

- set of vertex (nodes)
- set of edge (arcs)

each vertex pair connected by edge $\Rightarrow$ end points.



1. Trivial graph $\Rightarrow$. . .

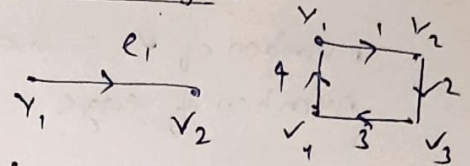
only 1 vertex
no edge.

2. Null graph $\Rightarrow$. . .

n vertex

no edge.

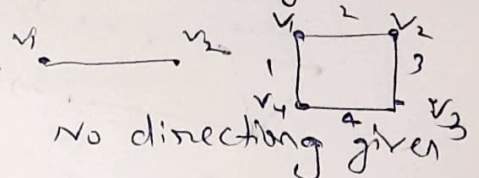
③ Directed graph



$G(\{v_1, v_2\}, \{e_1\})$

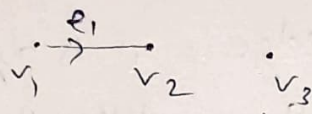
direction is given.

④ Undirected graph



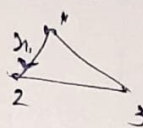
No direction given

5. Isolated vertex



not connected by 1 vertex of graph.

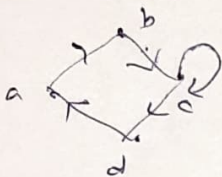
Mixed graph



direction +
no direction

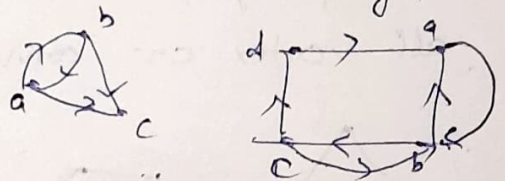
6. Self loop

containing loop at any vertex.



9. Multiple edge
(parallel edge)

Two vertex are connected
two or more edges.



7. Proper edge

without any loop.

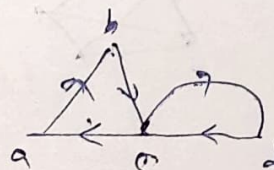


10. Simple graph

Containing
single structure



No loop, no multiple edge.



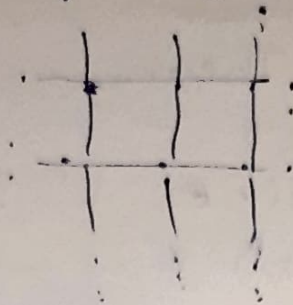
multi-graph

11. Finite graph



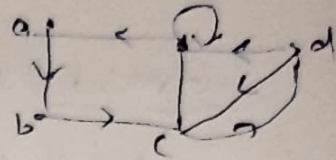
✓ both finite.

infinite graph



end points are not finite.
✓ both infinite.

12. Pseudo graph



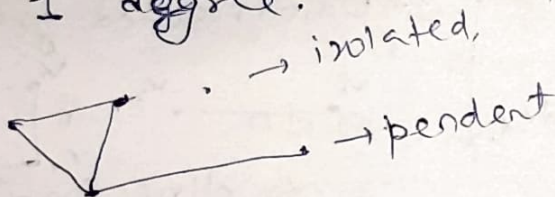
contains loop, multi-edge.

13. Order & Size

number of v = order
number of edge = size.

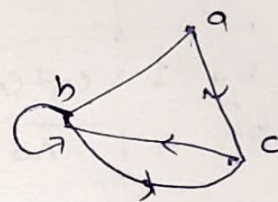
Pendent vertex

vertex which have 1 degree.



14. Degree

v vertex is connected to n edge = n is degree.



$$d(a) = 2$$

$$d(b) = 3$$

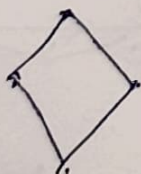
$$d(c) = 3$$

Isolated vertex,
degree = 0

Types of graph

1. Complete graph

Simple graph which all vertex are connected.



2. Regular graph

Simple graph in which all vertex have same degree = n

Note 3-regular graph
each degree = 3
for each v .

Unit 4 ~~5~~ } ~~5~~ ~~10~~ ~~15~~ ~~20~~ ~~25~~ ~~30~~ ~~35~~ ~~40~~ ~~45~~ ~~50~~ ~~55~~ ~~60~~ ~~65~~ ~~70~~ ~~75~~ ~~80~~ ~~85~~ ~~90~~ ~~95~~ ~~100~~ ~~105~~ ~~110~~ ~~115~~ ~~120~~ ~~125~~ ~~130~~ ~~135~~ ~~140~~ ~~145~~ ~~150~~ ~~155~~ ~~160~~ ~~165~~ ~~170~~ ~~175~~ ~~180~~ ~~185~~ ~~190~~ ~~195~~ ~~200~~ ~~205~~ ~~210~~ ~~215~~ ~~220~~ ~~225~~ ~~230~~ ~~235~~ ~~240~~ ~~245~~ ~~250~~ ~~255~~ ~~260~~ ~~265~~ ~~270~~ ~~275~~ ~~280~~ ~~285~~ ~~290~~ ~~295~~ ~~300~~ ~~305~~ ~~310~~ ~~315~~ ~~320~~ ~~325~~ ~~330~~ ~~335~~ ~~340~~ ~~345~~ ~~350~~ ~~355~~ ~~360~~ ~~365~~ ~~370~~ ~~375~~ ~~380~~ ~~385~~ ~~390~~ ~~395~~ ~~400~~ ~~405~~ ~~410~~ ~~415~~ ~~420~~ ~~425~~ ~~430~~ ~~435~~ ~~440~~ ~~445~~ ~~450~~ ~~455~~ ~~460~~ ~~465~~ ~~470~~ ~~475~~ ~~480~~ ~~485~~ ~~490~~ ~~495~~ ~~500~~ ~~505~~ ~~510~~ ~~515~~ ~~520~~ ~~525~~ ~~530~~ ~~535~~ ~~540~~ ~~545~~ ~~550~~ ~~555~~ ~~560~~ ~~565~~ ~~570~~ ~~575~~ ~~580~~ ~~585~~ ~~590~~ ~~595~~ ~~600~~ ~~605~~ ~~610~~ ~~615~~ ~~620~~ ~~625~~ ~~630~~ ~~635~~ ~~640~~ ~~645~~ ~~650~~ ~~655~~ ~~660~~ ~~665~~ ~~670~~ ~~675~~ ~~680~~ ~~685~~ ~~690~~ ~~695~~ ~~700~~ ~~705~~ ~~710~~ ~~715~~ ~~720~~ ~~725~~ ~~730~~ ~~735~~ ~~740~~ ~~745~~ ~~750~~ ~~755~~ ~~760~~ ~~765~~ ~~770~~ ~~775~~ ~~780~~ ~~785~~ ~~790~~ ~~795~~ ~~800~~ ~~805~~ ~~810~~ ~~815~~ ~~820~~ ~~825~~ ~~830~~ ~~835~~ ~~840~~ ~~845~~ ~~850~~ ~~855~~ ~~860~~ ~~865~~ ~~870~~ ~~875~~ ~~880~~ ~~885~~ ~~890~~ ~~895~~ ~~900~~ ~~905~~ ~~910~~ ~~915~~ ~~920~~ ~~925~~ ~~930~~ ~~935~~ ~~940~~ ~~945~~ ~~950~~ ~~955~~ ~~960~~ ~~965~~ ~~970~~ ~~975~~ ~~980~~ ~~985~~ ~~990~~ ~~995~~ ~~1000~~ ~~1005~~ ~~1010~~ ~~1015~~ ~~1020~~ ~~1025~~ ~~1030~~ ~~1035~~ ~~1040~~ ~~1045~~ ~~1050~~ ~~1055~~ ~~1060~~ ~~1065~~ ~~1070~~ ~~1075~~ ~~1080~~ ~~1085~~ ~~1090~~ ~~1095~~ ~~1100~~ ~~1105~~ ~~1110~~ ~~1115~~ ~~1120~~ ~~1125~~ ~~1130~~ ~~1135~~ ~~1140~~ ~~1145~~ ~~1150~~ ~~1155~~ ~~1160~~ ~~1165~~ ~~1170~~ ~~1175~~ ~~1180~~ ~~1185~~ ~~1190~~ ~~1195~~ ~~1200~~ ~~1205~~ ~~1210~~ ~~1215~~ ~~1220~~ ~~1225~~ ~~1230~~ ~~1235~~ ~~1240~~ ~~1245~~ ~~1250~~ ~~1255~~ ~~1260~~ ~~1265~~ ~~1270~~ ~~1275~~ ~~1280~~ ~~1285~~ ~~1290~~ ~~1295~~ ~~1300~~ ~~1305~~ ~~1310~~ ~~1315~~ ~~1320~~ ~~1325~~ ~~1330~~ ~~1335~~ ~~1340~~ ~~1345~~ ~~1350~~ ~~1355~~ ~~1360~~ ~~1365~~ ~~1370~~ ~~1375~~ ~~1380~~ ~~1385~~ ~~1390~~ ~~1395~~ ~~1400~~ ~~1405~~ ~~1410~~ ~~1415~~ ~~1420~~ ~~1425~~ ~~1430~~ ~~1435~~ ~~1440~~ ~~1445~~ ~~1450~~ ~~1455~~ ~~1460~~ ~~1465~~ ~~1470~~ ~~1475~~ ~~1480~~ ~~1485~~ ~~1490~~ ~~1495~~ ~~1500~~ ~~1505~~ ~~1510~~ ~~1515~~ ~~1520~~ ~~1525~~ ~~1530~~ ~~1535~~ ~~1540~~ ~~1545~~ ~~1550~~ ~~1555~~ ~~1560~~ ~~1565~~ ~~1570~~ ~~1575~~ ~~1580~~ ~~1585~~ ~~1590~~ ~~1595~~ ~~1600~~ ~~1605~~ ~~1610~~ ~~1615~~ ~~1620~~ ~~1625~~ ~~1630~~ ~~1635~~ ~~1640~~ ~~1645~~ ~~1650~~ ~~1655~~ ~~1660~~ ~~1665~~ ~~1670~~ ~~1675~~ ~~1680~~ ~~1685~~ ~~1690~~ ~~1695~~ ~~1700~~ ~~1705~~ ~~1710~~ ~~1715~~ ~~1720~~ ~~1725~~ ~~1730~~ ~~1735~~ ~~1740~~ ~~1745~~ ~~1750~~ ~~1755~~ ~~1760~~ ~~1765~~ ~~1770~~ ~~1775~~ ~~1780~~ ~~1785~~ ~~1790~~ ~~1795~~ ~~1800~~ ~~1805~~ ~~1810~~ ~~1815~~ ~~1820~~ ~~1825~~ ~~1830~~ ~~1835~~ ~~1840~~ ~~1845~~ ~~1850~~ ~~1855~~ ~~1860~~ ~~1865~~ ~~1870~~ ~~1875~~ ~~1880~~ ~~1885~~ ~~1890~~ ~~1895~~ ~~1900~~ ~~1905~~ ~~1910~~ ~~1915~~ ~~1920~~ ~~1925~~ ~~1930~~ ~~1935~~ ~~1940~~ ~~1945~~ ~~1950~~ ~~1955~~ ~~1960~~ ~~1965~~ ~~1970~~ ~~1975~~ ~~1980~~ ~~1985~~ ~~1990~~ ~~1995~~ ~~2000~~ ~~2005~~ ~~2010~~ ~~2015~~ ~~2020~~ ~~2025~~ ~~2030~~ ~~2035~~ ~~2040~~ ~~2045~~ ~~2050~~ ~~2055~~ ~~2060~~ ~~2065~~ ~~2070~~ ~~2075~~ ~~2080~~ ~~2085~~ ~~2090~~ ~~2095~~ ~~2100~~ ~~2105~~ ~~2110~~ ~~2115~~ ~~2120~~ ~~2125~~ ~~2130~~ ~~2135~~ ~~2140~~ ~~2145~~ ~~2150~~

3. Bigraph (Bipartite)

$$V_i = \{v_1, v_2, v_3, \dots, v_6\}$$

$$X = \{v_1, v_3, v_5\}$$

$$Y = \{v_2, v_4, v_6\}$$

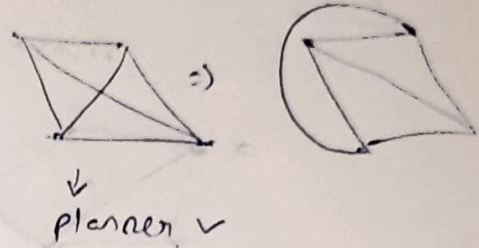
$$V = XUY$$

X & Y are decided in

non empty subset, x has 1
end & y has one end.

4. Plenen graph

If it can be drawn
with intersecting any
edge.



5. Isomorphism graph

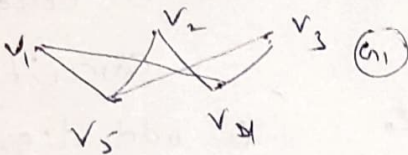
two graph

$$G_1 = (N_1, E_1)$$

$$C_2 = (v_2, e_2)$$

bijection mapping b/w

51 & 52



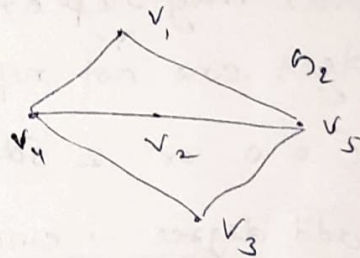
$$f: V \rightarrow V_2$$

- * Same number of vortex.

- Same number of edges

- degree sequence same.

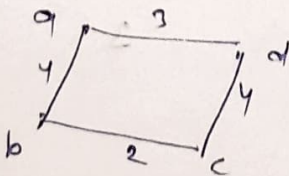
* similar structure



iso m o p h i s m

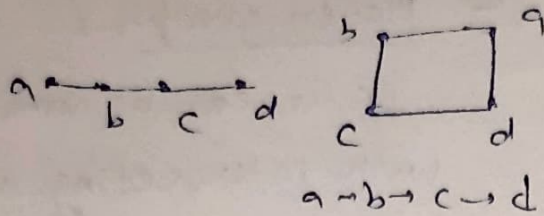
(b) weighted graph

each edge is assigned a real value



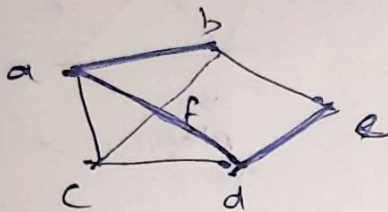
$$e(4, \alpha) = 3$$

walk all vertices are connected in following preceding way.



$(V_0 \text{ to } V_n)$

Path a walk in which no vertex appears more than 1 time. $V_0 \neq V_n$



$b \rightarrow a \rightarrow f \rightarrow d \rightarrow e$

length $\Rightarrow$ number of edge.

Cycle closed path + no vertex repeated.

Eulerian path

- vertex may repeat
- edges are not repeated.

degree $\equiv 0$ or 2 odd degree.

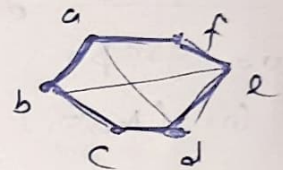
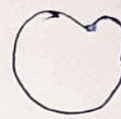
0-odd degree $\Rightarrow$ circuit

2-odd degree $\rightarrow$ path.

more than 2 $\rightarrow$ No eulerian.

Circuit closed path.

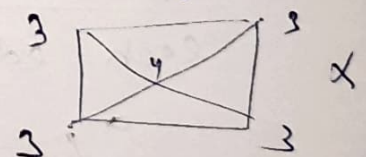
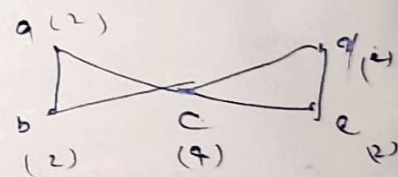
no edge appear
more than 1 time



Eulerian graph

Contains Eulerian circuit.

0-odd degree edge.

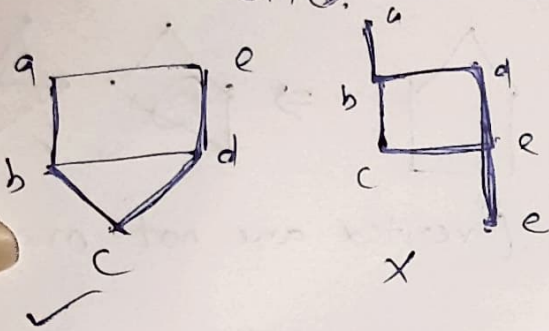


Hamilton path

vertex not repeated
edge may repeat
(visit every vertex once)

Hamilton circuit

start & end with same vertex.



each vertex has

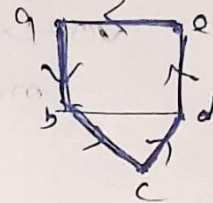
$$\text{degree} \geq n/2$$

K_n = always Hamiltonian.

↓
complete graph.

Hamilton graph

contains Hamilton cycle.
start & end same point.



K_n $n \geq 3$ Hamiltonian.

$$\sum (\text{deg}(v)) \geq n (\text{vertex})$$

* graph coloring

No two adjacent vertices have same color.

neighbours have diff color.

Chromatic Number.

minimum number of colors needed to properly color a graph

Type → face (plane)
↓
Vertex edge

Isolated = 1

Bipartite = 2

Complete graph = n

odd cycle (Chromatic number)

even cycle = 2

odd cycle = 3

triangle

($n \geq 3$)

$$\chi(G) \leq \Delta(G) + 1$$

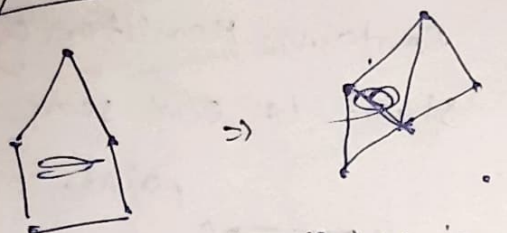
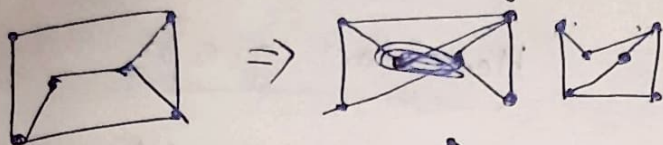
$\Delta(G)$ = maximum degree

Isomorphism

bijection.

$$f: v_1 \rightarrow v_2$$

same structure, diff. drawing.



vertex is moved.

- same vertices
- same edges
- same degree sequence
- same connectivity

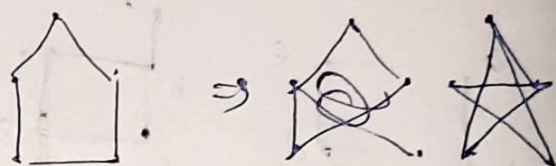
Homomorphism

$$f: V(G_1) \rightarrow V(G_2)$$

- Structure preserved partially

- Adjacent vertex remain adjacent.

no bijective.

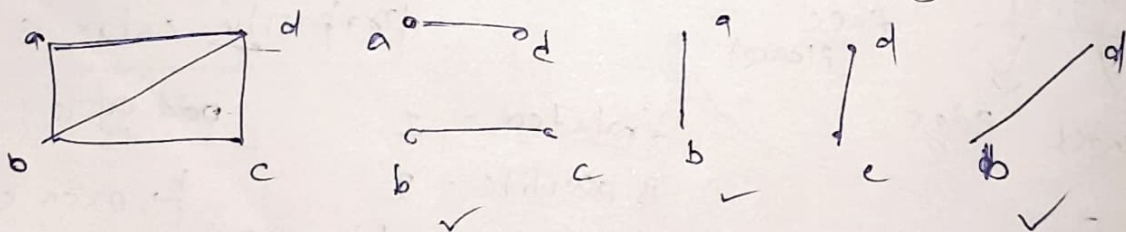


(vertices are not moved)

- No same vertices.
- No same number of edge.
- edges are divided into parts.

Matching graph

Sub graph of Graph, ^{edge} Sharing a common vertex.
no vertex is used more than once



Maximum Matching

matching with largest number of edge.

Perfect matching

every vertex matched
(Covered exactly 2)